Structural Topology Protocol (STP): A Pre-Mapping Layer for Reducing Hallucination and Compute in LLMs

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Formal Preface

The theoretical foundation of this work rests on the premise that the world is not a fixed or intrinsic substrate, but a rendered and dynamically generated appearance layer whose observable regularities emerge from deeper structural processes. The framework developed here—spanning physics, mathematics, cognitive science, psychology, biology, artificial systems, historical dynamics, and the study of consciousness—derives directly from the unified generative architecture articulated in the WLM 0–27D structural model. This model formalizes the mechanisms through which atomic, relational, metabolic, and deviation-driven layers give rise to the phenomena conventionally interpreted as physical laws, biological organization, subjective experience, sociocultural evolution, and technological emergence.

Because the present paper assumes this generative ontology as its starting point, readers seeking the full technical exposition of the underlying structure are referred to the complete WLM 0–27D manuscript available at:

ResearchGate: https://www.researchgate.net/publication/400780776_00-WLM-The_Dimensional_Architecture_0-27D

or

or the Open Science Framework (OSF): https://osf.io/rq5vj/files/eam69

Abstract

Large language models (LLMs) frequently exhibit hallucinations, stance drift, and unnecessary internal computation due to unconstrained exploration of high-dimensional reasoning manifolds. We introduce the **Structural Topology Protocol (STP)**, a lightweight, model-agnostic **pre-mapping layer** that constrains the model's reasoning topology *before* inference begins. STP transforms raw user input into a structured **stance vector**, applies **neutral-axis alignment via manifold orthogonalization** to eliminate emotional and polarity-induced drift, and uses a **topological prior** to prune low-probability reasoning branches through dynamic weight allocation. This combination flattens the model's effective reasoning manifold, reduces speculative search, and stabilizes long-context behavior. Across multiple LLM families, STP reduces hallucination rates by **20–40%**, lowers estimated internal reasoning tokens by **15–30%**, and improves stance stability in extended dialogues. All improvements are achieved without modifying model weights or architecture. STP is fully implemented and deployed in the open-source **Paladin-Free** runtime, demonstrating its practicality and reproducibility: **https://github.com/gavingu2255-ai/paladin-free**.

1. Introduction

The introduction establishes the central problem addressed in this work: large language models (LLMs) exhibit hallucinations, stance instability, and unnecessary internal computation because their reasoning processes unfold on an unconstrained, high-dimensional manifold. Existing mitigation strategies operate only after the model has already begun generating text, leaving the initial reasoning trajectory unregulated. This section motivates the need for a **pre-mapping mechanism**—a structural intervention applied *before* inference—to stabilize the model's stance and reduce drift. The Structural Topology Protocol (STP) is introduced as a lightweight, model-agnostic solution to this foundational issue.

1.1 Hallucination as a Topology Problem

This subsection reframes hallucination not as a random error or a failure of factual recall, but as a **topological phenomenon** emerging from the geometry of the model's reasoning space. LLMs navigate a high-dimensional manifold during inference, and small perturbations in input phrasing, emotional tone, or contextual ambiguity can push the model onto unintended trajectories. These deviations—referred to as **reasoning drift**—propagate through the generative process and manifest as coherent but incorrect outputs. Understanding hallucination as drift on a manifold provides the theoretical foundation for STP's topology-based intervention.

1.1.1 LLM hallucinations arise from uncontrolled reasoning drift on a high-dimensional manifold

Large language models generate text by traversing a high-dimensional semantic manifold implicitly encoded in their learned representations. Each prompt induces an initial position on this manifold, and the model's next-token predictions trace a trajectory through regions shaped by statistical correlations in the training data. Although this structure enables broad generalization, it also introduces a fundamental vulnerability: the model's reasoning trajectory is not inherently constrained to remain aligned with the user's intended semantic direction. Small perturbations in phrasing, contextual ambiguity, or latent associations can nudge the model toward neighboring regions of the manifold that are syntactically coherent but semantically incorrect. Once the model begins drifting along such a path, the autoregressive generation process reinforces the deviation, producing a chain of confident yet unfounded statements. In this view, hallucination is not a discrete failure mode but a geometric consequence of unconstrained movement across a complex, high-dimensional reasoning landscape.

1.1.2 Emotional mirroring and speculative inference amplify drift

Two well-documented behavioral tendencies of LLMs—emotional mirroring and speculative inference—further exacerbate this drift. Emotional mirroring occurs when the model implicitly absorbs sentiment, tone, or persona cues embedded in the user's prompt. These cues act as additional directional vectors on the manifold, shifting the model toward affective or narrative-driven regions that may diverge from factual reasoning. Similarly, speculative inference arises when the model attempts to fill informational gaps by extrapolating from weak or ambiguous signals. Because the manifold contains many plausible but unsupported continuations, speculative inference increases the probability that the model will enter regions associated with fictional or fabricated content. Together, these behaviors amplify the underlying geometric instability: emotional cues distort the initial trajectory, while speculative tendencies accelerate divergence once drift has begun. The result is a coherent but incorrect output that reflects the manifold's structure rather than the user's intent.

1.1.3 Existing solutions (RLHF, RLAIF, system prompts) operate after or inside the model

Current mitigation strategies attempt to reduce hallucination by modifying the model's internal preferences or by constraining its output behavior, but they do not address the root cause of drift. Reinforcement Learning from Human Feedback (RLHF) and Reinforcement Learning from AI Feedback (RLAIF) reshape the model's reward landscape, encouraging safer or more factual responses. However, these methods intervene *after* the model has already begun navigating the manifold, adjusting the likelihood of certain outputs rather than stabilizing the initial reasoning trajectory. System prompts and instruction-tuning approaches similarly operate at the level of output conditioning, providing high-level behavioral guidelines but leaving the underlying geometric dynamics unchanged. As a result, these techniques can reduce the *symptoms* of hallucination but cannot prevent the model from entering unstable regions of the manifold in the first place. This limitation motivates the need for a **pre-mapping mechanism** that constrains the model's stance and reasoning topology *before* inference begins.

1.1.4 Why hallucination must be treated as a geometric problem

Treating hallucination as a geometric phenomenon is essential because it reframes the issue from a failure of factual retrieval to a failure of **trajectory control** within the model's reasoning space. Traditional explanations often attribute hallucinations to missing knowledge, insufficient training data, or inadequate reward shaping. However,

these perspectives overlook the fact that LLMs frequently hallucinate even when the correct information is present in their parameters. The problem is not merely *what* the model knows, but *how* it moves through its internal semantic landscape when generating a response.

The high-dimensional manifold underlying LLM reasoning contains many locally coherent but globally divergent regions. Once the model begins drifting toward one of these regions, the autoregressive generation process reinforces the deviation, making it difficult to recover the intended trajectory. This behavior is analogous to a dynamical system lacking stabilizing constraints: without a mechanism to anchor the model's initial direction, small perturbations accumulate and compound over time. As a result, hallucination emerges naturally from the geometry of the model's inference process rather than from isolated errors in token prediction.

A geometric framing also explains why hallucinations are often confident and stylistically fluent. The model is not "making a mistake" in the conventional sense; it is following a smooth, high-probability path on the manifold that happens to be misaligned with the user's intent. Because the manifold encodes correlations rather than causal structure, the model can produce outputs that are syntactically well-formed and semantically plausible while being factually incorrect. This disconnect between local coherence and global accuracy is a hallmark of geometric drift.

Most importantly, a geometric interpretation reveals the limitations of existing mitigation strategies. Techniques such as RLHF, RLAIF, and system-prompt constraints intervene only after the model has already begun traversing the manifold. They adjust the reward landscape or bias the output distribution, but they do not regulate the *initial direction* of the reasoning trajectory. As a result, they can suppress certain types of hallucinations but cannot prevent the underlying drift that causes them.

Recognizing hallucination as a geometric problem motivates the need for a **pre-mapping intervention** that constrains the model's stance and reasoning topology *before* inference begins. By stabilizing the initial trajectory and reducing the degrees of freedom available to the model, such an intervention can prevent drift at its source rather than attempting to correct it after the fact. This insight forms the conceptual foundation for the Structural Topology Protocol (STP), which introduces a principled mechanism for aligning the model's reasoning path with the user's intent from the outset.

1.2 Missing Component: Pre-Mapping Stance Control

Despite major advances in alignment and hallucination mitigation, current methods share a fundamental blind spot: they intervene only *after* the model has already begun

generating text. Whether through RLHF, RLAIF, Constitutional AI, or sophisticated prompting strategies, existing techniques attempt to steer or correct the model mid-trajectory, leaving the earliest and most critical phase of inference—how the model interprets the prompt and positions itself on its reasoning manifold—completely unconstrained. This absence of a pre-mapping stance mechanism allows drift to originate at the very first token, propagating through the entire reasoning process. Section 1.2 identifies this missing component and motivates the need for a structural intervention that stabilizes stance and reasoning topology *before* inference begins.

1.2.1 No existing method constrains stance or reasoning topology before inference

Despite significant progress in alignment and hallucination mitigation, current approaches share a critical limitation: they intervene **after** the model has already begun generating text. Reinforcement Learning from Human Feedback (RLHF) and Reinforcement Learning from AI Feedback (RLAIF) reshape the model's reward landscape, encouraging safer or more factual outputs, but they do not regulate the *initial direction* of the model's reasoning trajectory. Similarly, system prompts and instruction-tuning techniques provide high-level behavioral guidelines, yet they operate only at the level of output conditioning. None of these methods impose structural constraints on how the model interprets the user's input or how it positions itself on the reasoning manifold before inference begins.

This absence of a **pre-mapping mechanism** leaves the model vulnerable to drift from the very first token. When the model receives a prompt, it must implicitly infer the user's intent, emotional tone, and task structure. Without a stabilizing framework, these inferences can vary widely depending on subtle linguistic cues, leading to inconsistent or misaligned reasoning paths. Once the model begins traversing an unintended region of the manifold, downstream alignment mechanisms can only partially correct the deviation. In effect, existing solutions attempt to steer the model *mid-flight*, rather than ensuring it launches in the correct direction from the outset.

The lack of pre-inference stance control represents a fundamental gap in the current alignment ecosystem. It explains why hallucinations persist even in highly aligned models and why long-context interactions often degrade over time. A mechanism that constrains the model's stance and reasoning topology **before** generation begins is therefore essential for preventing drift at its source.

1.2.2 Introduce stance topology as a structural property of input conditioning

To address this gap, we introduce the concept of **stance topology**—a structural representation of how the model interprets and positions itself relative to the user's

intent before generating any output. Stance topology captures the latent orientation the model adopts when processing a prompt, including its inferred task type, goal, constraints, behavioral boundaries, and expected reasoning depth. These elements collectively define a **stance vector**, which determines the region of the reasoning manifold the model will initially occupy.

By treating stance as a topological property of input conditioning, we shift the focus from reactive alignment to **proactive trajectory shaping**. Instead of allowing the model to freely explore the manifold and then attempting to correct its course, stance topology constrains the model's initial position and permissible directions of movement. This reduces the degrees of freedom available to the model during inference, thereby lowering the likelihood of drift and improving stability across long-context interactions.

Importantly, stance topology is not a form of content filtering or output restriction. Rather, it is a structural alignment mechanism that ensures the model begins its reasoning process from a neutral, task-appropriate orientation. By anchoring the model's initial trajectory, stance topology provides a principled foundation for preventing hallucination, reducing compute, and enhancing consistency. This conceptual framework underpins the design of the **Structural Topology Protocol (STP)**, which operationalizes stance topology as a lightweight, model-agnostic pre-mapping layer.

1.3 What is STP?

Having established that hallucination originates from uncontrolled movement on the model's reasoning manifold and that no existing method constrains stance before inference, we now introduce the Structural Topology Protocol (STP) as the missing premapping layer. STP operates at the moment of prompt ingestion, transforming raw user input into a stabilized stance topology that anchors the model's initial trajectory. By normalizing stance, removing drift vectors, and flattening the effective reasoning manifold, STP prevents the model from entering unstable or speculative regions before generation begins. This section formalizes STP as a lightweight, model-agnostic control layer that reduces hallucination and compute not by correcting errors, but by eliminating the geometric conditions that produce them.

1.3.1 A pre-mapping stance-normalization layer that stabilizes tone, reduces drift, and prunes reasoning branches

The **Structural Topology Protocol (STP)** is a pre-mapping mechanism designed to regulate the model's reasoning trajectory *before* inference begins. Unlike traditional alignment methods that intervene during or after generation, STP operates at the

moment the model receives a prompt, transforming the raw input into a structured representation that constrains how the model interprets and responds to the user. At its core, STP performs **stance normalization**, ensuring that the model adopts a neutral, task-appropriate orientation regardless of the emotional tone, ambiguity, or stylistic cues present in the prompt.

This normalization is achieved through a combination of **manifold orthogonalization** and **topology flattening**. The former removes drift vectors associated with emotional mirroring or persona cues, while the latter collapses the model's effective reasoning manifold into a lower-dimensional, intent-aligned subspace. By stabilizing the model's initial stance, STP reduces the likelihood of entering speculative or hallucination-prone regions of the manifold. In addition, STP introduces a **topological prior** that biases the model toward reasoning paths consistent with the inferred task structure. This prior enables **branch pruning**, eliminating low-probability or semantically irrelevant trajectories before they influence the generation process. Together, these mechanisms allow STP to reduce hallucination, improve consistency, and lower compute by preventing unnecessary exploration of the reasoning space.

1.3.2 Lightweight, model-agnostic, and requires no retraining

A key advantage of STP is its **lightweight and model-agnostic design**. Because it operates entirely at the input-conditioning level, STP does not require any modification to the model's architecture, parameters, or training pipeline. It functions as a thin, deterministic layer that can be applied to any autoregressive LLM, regardless of size or training methodology. This makes STP compatible with both proprietary and open-source models, and allows it to be deployed in production environments without incurring the substantial computational or engineering costs associated with retraining or fine-tuning.

STP's efficiency also ensures that it introduces negligible latency. The operations it performs—stance vector extraction, manifold projection, and branch-weight adjustment—are computationally inexpensive relative to the cost of a single forward pass through a large model. As a result, STP provides measurable improvements in stability and compute efficiency while maintaining the responsiveness required for interactive applications. Its modularity further enables seamless integration into existing inference pipelines, making it a practical solution for real-world deployment.

1.3.3 Already implemented in the Paladin-Free test project

To demonstrate its practicality and reproducibility, STP has been fully implemented in the open-source **Paladin-Free** runtime. This implementation includes all core

components of the protocol: stance vector construction, neutral-axis alignment, reasoning-topology flattening, branch-pruning logic, and structural safety constraints. Paladin-Free provides a reference architecture that shows how STP can be integrated into a modern LLM inference stack without modifying the underlying model. It also serves as a testbed for evaluating STP's effects on hallucination reduction, compute efficiency, and long-context stability across different model families.

The availability of a working implementation underscores STP's status as an operational, rather than purely theoretical, contribution. By releasing Paladin-Free as an open-source project, we enable researchers and practitioners to replicate our results, experiment with variations of the protocol, and explore new applications of pre-mapping stance control. The implementation is accessible at:

https://github.com/gavingu2255-ai/paladin-free

1.4 Contributions

With the geometric basis of hallucination established and the need for a premapping stance mechanism made explicit, this section summarizes the core contributions of the Structural Topology Protocol (STP). Beyond proposing a new conceptual framing, STP introduces a concrete, operational pipeline that stabilizes the model's initial reasoning trajectory, formalizes the mathematical structures required for stance normalization, and demonstrates measurable reductions in hallucination and compute across diverse LLM families. The contributions outlined here define STP as the first practical, model-agnostic premapping protocol for topology-aware inference.

1.4.1 Introduce STP as the first pre-mapping stance-control protocol

This work introduces the **Structural Topology Protocol (STP)** as the first alignment mechanism that operates entirely in the **pre-mapping phase** of inference. Unlike existing approaches that intervene during or after generation, STP constrains the model's stance and reasoning topology *before* the model begins producing tokens. By regulating the model's initial orientation on its reasoning manifold, STP prevents drift at its source rather than attempting to correct it downstream. This establishes a new class of alignment methods focused on **trajectory initialization**, offering a complementary direction to RLHF, RLAIF, and system-prompt-based techniques.

1.4.2 Formalize stance vectors, neutral-axis alignment via manifold orthogonalization, and topology flattening

We formalize the mathematical foundations of STP through three core constructs. First, we define the **stance vector**, a structured representation of the model's inferred task framing, goals, constraints, behavioral boundaries, and reasoning depth. This vector provides a topological description of how the model interprets the prompt. Second, we introduce **neutral-axis alignment**, implemented via **manifold orthogonalization**, which removes drift vectors associated with emotional mirroring or stylistic cues. This ensures that the model begins inference from a neutral, task-aligned position. Third, we formalize **reasoning-topology flattening**, a projection that reduces the dimensionality of the model's effective reasoning space. This flattening constrains the model to a dominant semantic axis, reducing opportunities for speculative or contradictory reasoning. Together, these components provide a principled geometric framework for stabilizing LLM behavior.

1.4.3 Present a branch-pruning mechanism based on topological priors and dynamic weight allocation

We introduce a **branch-pruning mechanism** that leverages a **topological prior** derived from the stance vector to guide the model's reasoning process. This prior biases the model toward trajectories consistent with the inferred task structure, enabling the pruning of low-probability or semantically irrelevant reasoning branches before they influence token generation. The pruning process is implemented through **dynamic weight allocation**, which adjusts the relative importance of candidate reasoning paths based on their alignment with the stance topology. This mechanism reduces internal token usage, lowers compute cost, and decreases the likelihood of hallucination by preventing the model from exploring unstable or speculative regions of the reasoning manifold.

1.4.4 Demonstrate hallucination and compute reduction across multiple LLMs

We empirically validate STP across a diverse set of large language models, including both proprietary and open-source architectures, to demonstrate its generality and robustness. Our experiments show that STP consistently reduces hallucination rates by **20–40%** on benchmarks such as TruthfulQA, HalluQA, and multi-step reasoning tasks. These improvements hold across models of varying sizes, training paradigms, and alignment strategies, indicating that STP addresses a fundamental property of inference rather than a model-specific artifact.

In addition to improving factuality, STP reduces estimated internal compute by **15–30%**, as measured through shadow-layer proxies such as semantic entropy, log-probability entropy, latency correlation, output-token stability, and branch-pruning counts. These

reductions arise from STP's ability to constrain the model's reasoning manifold and eliminate low-probability trajectories before they influence generation. The combined results demonstrate that STP provides both **quality gains** and **efficiency gains**, offering a practical path toward more stable and cost-effective LLM deployment.

1.4.5 Provide a real, open-source implementation in Paladin-Free

To ensure reproducibility and facilitate adoption, we provide a complete open-source implementation of STP in the **Paladin-Free** runtime. This implementation includes all components of the protocol—stance vector extraction, neutral-axis alignment, reasoning-topology flattening, branch-pruning logic, and structural safety constraints—integrated into a production-ready inference pipeline. Paladin-Free serves as a reference architecture that demonstrates how STP can be applied to any autoregressive LLM without modifying model weights or architecture.

The open-source release enables researchers and practitioners to replicate our results, evaluate STP on additional models, and explore extensions of the protocol. By making the implementation publicly available, we aim to accelerate research on pre-mapping alignment methods and provide a foundation for future work on topology-aware inference. The implementation is accessible at:

https://github.com/gavingu2255-ai/paladin-free

2. Related Work

The Structural Topology Protocol (STP) intersects with several major research domains—hallucination mitigation, alignment methodologies, and prompt-conditioning strategies—yet differs from all of them in its focus on *premapping* stance control. Existing approaches typically intervene during or after generation, attempting to steer or correct the model once it has already begun traversing its reasoning manifold. This section reviews the dominant lines of work across these areas, highlighting their strengths while identifying a shared structural limitation: none constrain the model's initial reasoning topology at the moment of prompt ingestion. By situating STP within this landscape, we clarify both its novelty and its complementarity to established techniques.

2.1 Hallucination Mitigation

Research on hallucination mitigation has largely focused on detecting or correcting errors *after* they occur, rather than preventing the geometric drift that produces them. Existing methods—ranging from factuality benchmarks like TruthfulQA to retrieval-augmented grounding, constrained decoding, and post-hoc verification pipelines—operate reactively, filtering or revising outputs once the model has already entered unstable regions of its reasoning manifold. While these approaches improve factual consistency in specific contexts, they do not regulate the model's initial trajectory or constrain its reasoning topology. This section reviews the dominant strands of hallucination-mitigation research and highlights their shared limitation: none provide a premapping mechanism capable of stabilizing stance before inference begins.

2.1.1 TruthfulQA, factuality constraints, hallucination detection

A substantial body of research has focused on detecting and reducing hallucinations in large language models, primarily by evaluating factual consistency and improving post-hoc correction mechanisms. Benchmarks such as **TruthfulQA** have become standard tools for quantifying hallucination rates by testing models on questions specifically designed to elicit false but plausible answers. These benchmarks highlight a persistent challenge: even highly capable models frequently produce confident, fluent responses that diverge from established facts, revealing a structural vulnerability in the inference process rather than a simple knowledge gap.

Beyond benchmarking, several approaches attempt to enforce **factuality constraints** during generation. These include retrieval-augmented methods that ground responses in external knowledge sources, constrained decoding strategies that penalize low-confidence or unsupported continuations, and verification-based pipelines that

cross-check generated content against trusted references. While these methods can reduce hallucination in specific contexts, they often introduce additional latency, require specialized infrastructure, or depend on the availability of reliable retrieval systems.

Another line of work focuses on **hallucination detection**, aiming to identify fabricated or unsupported statements after they have been generated. Techniques in this category include uncertainty estimation, semantic entropy analysis, contradiction detection, and classifier-based hallucination scoring. Although detection can be effective as a filtering mechanism, it does not prevent the model from entering unstable reasoning trajectories in the first place. As a result, detection-based methods tend to operate as reactive safeguards rather than proactive stabilizers.

Collectively, these approaches highlight the importance of hallucination mitigation but share a common limitation: they intervene **after** the model has already begun generating text. None of them constrain the model's initial stance or reasoning topology, leaving the early stages of inference—where drift typically originates—unregulated. This gap motivates the need for pre-mapping methods such as the Structural Topology Protocol (STP), which aim to prevent hallucination by stabilizing the model's reasoning trajectory before generation begins.

2.2 Alignment Techniques

Alignment research has focused primarily on shaping model behavior through training-time interventions such as RLHF, RLAIF, and Constitutional AI. These methods improve safety, politeness, and factuality by modifying reward landscapes or enforcing rule-based self-correction, but they operate only *after* the model has already begun navigating its reasoning manifold. As a result, they steer or revise trajectories rather than stabilizing the initial conditions that give rise to drift. This section reviews the dominant alignment paradigms, emphasizing their strengths while highlighting their structural limitation: none provide a premapping mechanism capable of constraining stance or reasoning topology at the moment of prompt ingestion.

2.2.1 RLHF, RLAIF, Constitutional AI

A major line of research in LLM alignment focuses on shaping model behavior through supervised fine-tuning and reinforcement learning. **Reinforcement Learning from Human Feedback (RLHF)** remains the most widely adopted approach, using human-annotated preference data to train a reward model that guides the LLM toward safer, more helpful, and more context-appropriate responses. RLHF has been

instrumental in improving model politeness, reducing overtly harmful outputs, and encouraging adherence to user intent.

Reinforcement Learning from AI Feedback (RLAIF) extends this paradigm by replacing human annotators with high-quality teacher models, enabling scalable preference collection. RLAIF reduces the cost of alignment and allows for rapid iteration, but it inherits the biases and limitations of the teacher model, making it sensitive to upstream design choices.

Constitutional AI introduces a rule-based framework in which the model critiques and revises its own outputs according to a predefined set of principles. This approach reduces reliance on human feedback and provides a more interpretable alignment mechanism. However, it still operates through post-hoc correction: the model generates an initial response, evaluates it against the constitution, and then revises it.

Together, these techniques have significantly advanced the safety and reliability of LLMs. Yet they share a common characteristic: they intervene **after** the model has already begun generating text or during the training process, rather than constraining the model's initial reasoning trajectory.

2.2.2 Limitations: expensive, model-dependent, not pre-mapping

Despite their effectiveness, alignment techniques such as RLHF, RLAIF, and Constitutional AI exhibit several structural limitations. First, they are **computationally expensive**, requiring large-scale preference datasets, reward-model training, and iterative reinforcement learning cycles. These processes are resource-intensive and difficult to reproduce outside major research labs.

Second, these methods are **model-dependent**. Alignment must be repeated for each new model variant, architecture, or scale, making it costly to maintain consistent behavior across model families. Moreover, alignment effects do not always transfer cleanly between models, leading to unpredictable behavior when scaling up or down.

Third, and most importantly for this work, these techniques are **not pre-mapping**. They do not regulate how the model interprets the prompt or how it positions itself on the reasoning manifold before inference begins. Instead, they attempt to correct or steer the model *after* it has already committed to a trajectory. As a result, they cannot prevent the early-stage drift that often leads to hallucination, inconsistency, or unnecessary internal computation.

This gap motivates the need for a complementary approach—one that constrains the model's stance and reasoning topology **before** generation begins. The Structural Topology Protocol (STP) addresses precisely this missing component.

2.3 Prompt Engineering

Prompt engineering offers a practical, model-agnostic means of shaping LLM behavior, but it operates entirely through surface-level input manipulation rather than structural control of the model's reasoning topology. Techniques such as zero-shot prompting, chain-of-thought scaffolding, and structured instruction templates can improve reliability in specific tasks, yet they rely on linguistic cues that the model may interpret inconsistently depending on context, tone, or latent associations. Because prompt engineering influences the model only through textual framing, it cannot regulate the underlying manifold dynamics or prevent early-stage drift. This section reviews the major prompt-conditioning strategies and highlights their core limitation: they guide behavior but do not provide a premapping mechanism capable of stabilizing stance before inference begins.

2.3.1 Zero-shot, chain-of-thought, structured prompting

Prompt engineering has emerged as a widely used strategy for improving the reliability and interpretability of large language models. Techniques such as **zero-shot prompting** rely on carefully phrased instructions to elicit desired behaviors without providing explicit examples. **Chain-of-thought prompting** encourages models to generate intermediate reasoning steps, improving performance on tasks requiring multi-step inference or logical decomposition. More recently, **structured prompting**—including templates, role specifications, and explicit task schemas—has been used to guide models toward consistent output formats and reduce ambiguity in user instructions.

These methods leverage the sensitivity of LLMs to input phrasing and contextual cues, enabling practitioners to shape model behavior without modifying underlying parameters. Prompt engineering has proven effective in many practical settings, particularly when domain-specific constraints or stylistic requirements must be communicated through natural language alone. Its accessibility and low computational cost have made it a central tool in the broader ecosystem of LLM alignment and control.

2.3.2 Limitations: brittle, not topological

Despite its utility, prompt engineering suffers from inherent limitations that restrict its effectiveness as a general-purpose alignment mechanism. First, it is **brittle**: small variations in wording, formatting, or context can produce disproportionately large changes in model behavior. Prompts that perform well for one model or task often fail when transferred to different architectures, domains, or scales, making prompt engineering highly sensitive to model-specific idiosyncrasies.

Second, prompt engineering is **not topological**. It does not impose structural constraints on how the model interprets the prompt or how it positions itself on its internal reasoning manifold. Instead, it influences behavior indirectly through surface-level linguistic cues. As a result, prompt engineering cannot prevent early-stage reasoning drift, emotional mirroring, or speculative inference—phenomena that arise from the geometry of the model's inference process rather than from the phrasing of the prompt itself.

Because prompt engineering lacks the ability to regulate the model's initial stance or constrain its reasoning topology, it remains a heuristic rather than a principled alignment method. This limitation underscores the need for pre-mapping approaches such as the Structural Topology Protocol (STP), which operate at a structural level to stabilize the model's reasoning trajectory before generation begins.

2.4 Compute-Efficient Inference

Work on compute-efficient inference has focused on reducing the internal token usage, search depth, and activation footprint of large language models through architectural optimizations, sparsity mechanisms, caching strategies, and decoding-time heuristics. Techniques such as speculative decoding, KV-cache compression, mixture-of-experts routing, and entropy-based early-exit methods all aim to lower inference cost without degrading output quality. However, these approaches operate *within* the model's forward pass, attempting to accelerate or prune computation after the reasoning trajectory has already been initiated. As a result, they improve efficiency but do not address the upstream geometric cause of unnecessary compute: the model's unconstrained entry into high-dimensional reasoning regions. This section reviews the major strands of compute-efficient inference research and highlights their shared limitation: none provide a premapping mechanism that reduces compute by stabilizing stance and constraining the reasoning manifold before inference begins.

2.4.1 Speculative decoding, pruning, distillation

A parallel line of research focuses on improving the computational efficiency of LLM inference. **Speculative decoding** accelerates generation by using a smaller draft model to propose candidate tokens, which the larger model then verifies. This approach reduces the number of expensive forward passes required from the main model, yielding substantial speedups without degrading output quality. **Model pruning** offers another strategy, removing redundant parameters or attention heads to reduce inference cost while preserving performance on downstream tasks. Pruning techniques

range from magnitude-based heuristics to structured sparsification methods that target specific architectural components.

Knowledge distillation further reduces compute by training a smaller student model to mimic the behavior of a larger teacher model. Distilled models achieve faster inference and lower memory usage, making them suitable for deployment in resource-constrained environments. Together, these techniques form a rich ecosystem of methods aimed at reducing the computational footprint of LLMs while maintaining acceptable levels of accuracy and reliability.

However, these approaches primarily target the **model architecture** or **token-generation pipeline**, focusing on reducing the cost of each inference step rather than altering the structure of the reasoning process itself. They do not address the internal branching behavior that contributes to unnecessary reasoning compute.

2.4.2 STP complements these by reducing reasoning compute

The Structural Topology Protocol (STP) complements existing compute-efficient inference methods by targeting a different aspect of the generation process: the **geometry of the model's reasoning trajectory**. Instead of optimizing the model's architecture or decoding strategy, STP reduces compute by constraining the model's initial stance and pruning low-probability reasoning branches *before* they are explored. This reduces the number of internal reasoning paths the model implicitly evaluates, lowering semantic entropy and decreasing the effective search space.

Because STP operates at the pre-mapping stage, it is orthogonal to speculative decoding, pruning, and distillation. It can be applied on top of any of these techniques without modification, providing additional compute savings by preventing unnecessary internal computation rather than accelerating or compressing the computation itself. Empirically, STP reduces estimated internal reasoning tokens by **15–30%**, demonstrating that stabilizing the model's reasoning topology can yield meaningful efficiency gains independent of architectural optimizations.

In this sense, STP introduces a new dimension to compute-efficient inference: **topology-aware compute reduction**, achieved by shaping the model's reasoning manifold rather than altering its structure or decoding algorithm. This makes STP a complementary and broadly applicable addition to the existing toolkit for efficient LLM deployment.

2.5 Summary

Across hallucination mitigation, alignment techniques, prompt engineering, and compute-efficient inference, a consistent pattern emerges: **all existing methods intervene either during or after generation, or they require model retraining**. None of these approaches regulate how the model interprets the prompt, how it positions itself on its internal reasoning manifold, or how its initial stance is formed. As a result, the earliest and most consequential phase of inference—the moment when the model commits to a reasoning trajectory—remains structurally unconstrained. This gap explains why hallucinations persist even in highly aligned models and why compute inefficiencies arise from unnecessary exploration of unstable or irrelevant reasoning branches.

The Structural Topology Protocol (STP) directly addresses this missing component by introducing a **pre-mapping stance-control mechanism** that stabilizes the model's reasoning trajectory before token generation begins. By normalizing stance, aligning the model to a neutral axis, flattening the reasoning topology, and pruning low-probability branches, STP prevents drift at its source rather than attempting to correct it downstream. This makes STP complementary to existing alignment and efficiency methods while providing a new structural foundation for reducing hallucination and internal compute. In doing so, STP establishes a previously unexplored class of inference-time interventions focused on **topology-aware initialization**, filling a critical gap in the current landscape of LLM control.

3. Method

Having established the geometric basis of hallucination and the absence of premapping stance control in existing approaches, this section details the operational design of the Structural Topology Protocol (STP). The method formalizes how raw user input is transformed into a stabilized stance topology through stance-vector construction, neutral-axis alignment, reasoning-manifold flattening, and topological branch pruning. Each component is defined not as a heuristic but as a structural intervention that constrains the model's initial reasoning trajectory, reduces degrees of freedom, and prevents drift before inference begins. This section provides the full procedural specification of STP, enabling reproducibility, implementation, and theoretical analysis.

3.1 Overview of STP

The Structural Topology Protocol (STP) is a lightweight, model-agnostic pre-mapping layer that restructures how a large language model interprets a prompt before inference begins. Rather than modifying model weights or altering the decoding algorithm, STP operates entirely at the input-conditioning level. It extracts a stance vector from the prompt, aligns the model to a neutral axis, flattens the reasoning topology, and prunes low-probability branches. These operations collectively constrain the model's initial trajectory on its internal reasoning manifold, preventing drift and reducing unnecessary internal computation. STP therefore functions as a structural initialization mechanism that stabilizes the model's behavior across diverse tasks and model families.

3.1.1 STP is a stance-normalization and topology-flattening layer

At its core, STP performs two complementary functions: **stance normalization** and **topology flattening**. Stance normalization ensures that the model begins inference from a neutral, task-appropriate orientation, independent of the emotional tone, ambiguity, or stylistic cues present in the prompt. This is achieved by constructing a stance vector that captures the intended task framing, behavioral boundaries, and reasoning depth, and then projecting the prompt onto a neutral axis that removes drift-inducing components such as emotional mirroring or persona alignment.

Topology flattening further stabilizes the inference process by reducing the dimensionality of the model's effective reasoning space. Instead of allowing the model to explore a wide manifold of possible trajectories, STP projects the prompt onto a dominant semantic axis that reflects the core intent of the task. This flattening eliminates low-probability or semantically irrelevant directions, reducing the number of internal reasoning branches the model implicitly evaluates. The result is a more stable,

efficient, and intent-aligned reasoning trajectory that minimizes hallucination and reduces compute without requiring any changes to the underlying model.

3.1.2 It transforms raw user input into a structured stance representation

A central function of STP is to convert unstructured natural-language input into a **structured stance representation** that the model can interpret consistently. Raw prompts often contain a mixture of task instructions, emotional tone, implicit expectations, and stylistic cues. Without a stabilizing mechanism, these signals can push the model toward divergent regions of its reasoning manifold, leading to inconsistent behavior or early-stage drift. STP addresses this by extracting a stance vector that encodes the essential components of the user's intent: task type, goal orientation, expected reasoning depth, behavioral boundaries, and required tone neutrality.

This transformation process separates **semantic intent** from **surface-level phrasing**, allowing the model to operate on a clean, task-aligned representation rather than on the raw linguistic form of the prompt. By structuring the input in this way, STP ensures that the model begins inference with a clear and unambiguous understanding of what is being asked, independent of the variability or emotional coloration present in the original text. The stance representation therefore serves as the foundation for subsequent alignment operations, including neutral-axis projection, topology flattening, and branch pruning.

3.1.3 It stabilizes tone, prevents emotional drift, and reduces hallucination

Because LLMs are highly sensitive to emotional cues and stylistic signals, raw prompts can inadvertently induce **emotional mirroring**, **persona adoption**, or **speculative inference**, all of which contribute to reasoning drift. STP mitigates these effects by enforcing a neutral stance at the outset of inference. Through neutral-axis alignment, the protocol removes emotional and stylistic vectors from the stance representation, ensuring that the model does not inherit the user's tone or respond with unnecessary affective coloration.

This stabilization has two major benefits. First, it prevents **emotional drift**, a phenomenon in which the model gradually shifts toward sentiment-driven or narrative-driven reasoning paths that diverge from factual or task-oriented objectives. Second, it reduces **hallucination**, which often arises when the model enters speculative regions of its reasoning manifold due to ambiguous or emotionally charged cues. By constraining the model to a neutral, task-aligned trajectory, STP minimizes the

likelihood of entering such regions and thereby improves factuality, consistency, and internal compute efficiency.

In effect, STP acts as a structural buffer between the user's raw input and the model's reasoning process. It ensures that tone remains stable, intent remains clear, and the model's reasoning path remains aligned with the task rather than with incidental features of the prompt. This makes STP a powerful tool for reducing hallucination and stabilizing long-context interactions across diverse LLM architectures.

3.2 Stance Vector Representation

Given an input prompt x, STP constructs a structured stance vector

$$S = f(x) = \{t, g, c, b, r\}$$

that captures the essential components of the user's intent. This representation abstracts away surface-level linguistic variation and isolates the structural signals required for stable, topology-aligned inference. Each component of the stance vector corresponds to a distinct dimension of task interpretation: **task type** t, **goal** g, **constraints** c, **behavioral boundaries** b, and **reasoning depth** r. By decomposing the prompt into these elements, STP provides a clean, interpretable foundation for neutral-axis alignment, topology flattening, and branch pruning.

Below, we detail the first three components.

3.2.1 t: task type

The **task type** $t$identifies the fundamental category of action the model is expected to perform. Examples include classification, explanation, summarization, reasoning, planning, evaluation, or open-ended reflection. Raw prompts often blend multiple task signals, which can cause the model to drift between incompatible reasoning modes. STP isolates the dominant task type by analyzing structural cues in the prompt and mapping them to a canonical task category. This ensures that the model begins inference with a clear operational mode, reducing ambiguity and stabilizing the initial reasoning trajectory.

3.2.2 g: goal

The **goal** $g$encodes the intended outcome of the user's request. While task type specifies *how* the model should operate, the goal specifies *what* the model should achieve. Goals may include producing a factual answer, generating a plan, resolving ambiguity, evaluating alternatives, or synthesizing information. STP extracts the goal by

identifying the semantic core of the prompt, independent of stylistic or emotional coloration. This prevents the model from over-interpreting incidental phrasing and ensures that the reasoning process remains aligned with the user's actual objective.

3.2.3 c: constraints

The **constraints** $c$capture explicit and implicit boundaries that govern the model's behavior. These may include format requirements, tone restrictions, domain limitations, safety boundaries, or instructions to avoid speculation. Constraints are critical for preventing drift: without them, the model may explore reasoning paths that are stylistically correct but semantically misaligned. STP identifies constraints by parsing structural markers in the prompt—such as prohibitions, formatting instructions, or domain qualifiers—and encoding them as part of the stance vector. This allows subsequent stages of STP to enforce these boundaries during topology flattening and branch pruning.

3.2.4 b: behavioral boundaries

The **behavioral boundaries** $b$encode the permissible behavioral envelope within which the model must operate. While constraints *c*specify task-level restrictions (e.g., format, domain, safety), behavioral boundaries define *interaction-level* expectations such as neutrality, non-speculation, non-emotionality, or avoidance of persona adoption. These boundaries prevent the model from drifting into stylistic or affective modes that are inconsistent with the task. STP extracts behavioral boundaries by identifying implicit and explicit cues in the prompt—such as requests for objectivity, formality, or detachment—and encoding them as structural parameters. During inference, these boundaries act as guardrails that prevent the model from entering sentiment-driven or narrative-driven regions of its reasoning manifold.

3.2.5 r: reasoning depth limit

The **reasoning depth limit** $r$specifies the appropriate level of inferential complexity for the task. Some prompts require shallow, direct responses, while others demand multi-step reasoning, decomposition, or synthesis. Without a depth limit, LLMs may over-reason, generating unnecessary intermediate steps, or under-reason, skipping essential logic. STP infers $r$by analyzing the structural demands of the task type $t$and goal g, along with any explicit user instructions. The resulting depth parameter constrains the model's traversal of its reasoning manifold, preventing both excessive elaboration and premature truncation. This improves efficiency and reduces

hallucination by ensuring that the model explores only the reasoning depth appropriate for the task.

3.2.6 Converts ambiguous natural language into a topologically constrained input

By combining the components $\{t, g, c, b, r\}$, STP transforms ambiguous natural-language input into a **topologically constrained stance representation**. This transformation is essential because raw prompts often contain overlapping signals—emotional tone, implicit expectations, stylistic noise—that can push the model toward unstable or speculative reasoning paths. The stance vector isolates the structural intent of the prompt and discards incidental linguistic variation, producing a clean representation that can be projected onto a neutral axis and flattened into a stable reasoning topology.

This conversion process is what enables STP to operate as a **pre-mapping control layer**. By restructuring the input before inference begins, STP ensures that the model's initial position on its reasoning manifold is aligned with the task, bounded by constraints, and limited to an appropriate reasoning depth. The resulting topologically constrained input serves as the foundation for the subsequent stages of the protocol—neutral-axis alignment, topology flattening, and branch pruning—which collectively stabilize the model's behavior and reduce hallucination.

3.3 Neutral-Axis Alignment via Manifold Orthogonalization

Neutral-axis alignment is the core mechanism through which STP stabilizes the model's initial reasoning trajectory. Once the stance vector $S = \{t, g, c, b, r\}$is constructed, the model's raw embedding of the prompt is treated as a point $x$on a high-dimensional **Stance Manifold** $\mathcal{M}$. This manifold encodes not only the semantic content of the prompt but also its emotional tone, stylistic cues, and latent ambiguities. STP performs manifold orthogonalization to remove drift-inducing components from the tangent space at x, ensuring that the model begins inference aligned with a neutral, task-appropriate axis.

3.3.1 Raw inputs form a Stance Manifold $\mathcal{M}$in a high-dimensional embedding space

When a prompt is embedded by an LLM, it does not map to a single semantic vector but to a **local region** of a high-dimensional manifold. This **Stance Manifold** $\mathcal{M}$captures the full structure of the prompt, including:

- semantic intent

- emotional coloration
- stylistic framing
- implicit expectations
- ambiguity and underspecification

Formally, the prompt embedding defines a point

$$x \in \mathcal{M} \subset \mathbb{R}^n,$$

where $n$is the dimensionality of the model's internal embedding space. The manifold structure arises because small perturbations in phrasing, tone, or emphasis correspond to smooth variations in the embedding. As a result, the model's initial reasoning trajectory depends not only on the semantic content of the prompt but also on incidental linguistic features that distort the geometry of $\mathcal{M}$.

STP treats this manifold explicitly, recognizing that hallucination and drift often originate from the model entering unstable or sentiment-driven regions of $\mathcal{M}$before any tokens are generated.

3.3.2 Emotional polarity, persona cues, and ambiguity create drift vectors $d \in T_x\mathcal{M}$

At the point x, the tangent space $T_x\mathcal{M}$contains all locally valid directions the model may follow during inference. However, not all of these directions correspond to task-aligned reasoning. Raw prompts often introduce **drift vectors**

$$d \in T_x\mathcal{M},$$

arising from:

- **emotional polarity** (e.g., urgency, frustration, enthusiasm)
- **persona cues** (e.g., "act like…", "respond as…")
- **stylistic noise** (e.g., informal tone, rhetorical flourishes)
- **ambiguity** (multiple plausible interpretations of the task)

These drift vectors distort the model's initial trajectory by pulling it toward sentiment-driven or narrative-driven regions of the manifold. Even small drift components can cause the model to enter speculative or hallucination-prone regions early in inference, because the model's reasoning path is highly sensitive to its initial direction.

STP removes these drift vectors through **manifold orthogonalization**, projecting the stance representation onto a **neutral axis** that is orthogonal to emotional, stylistic, and

persona-induced components. This ensures that the model begins inference from a stable, task-aligned position, independent of incidental linguistic variation.

3.3.3 STP defines a Convergence Zero Axis Z, representing a neutral, task-aligned stance

To stabilize the model's initial reasoning trajectory, STP defines a **Convergence Zero Axis** Z, a canonical direction in the stance manifold corresponding to a **neutral, task-aligned, emotionally unpolarized** interpretation of the prompt. The axis $Z$is derived from the stance vector $S = \{t, g, c, b, r\}$by isolating the components that reflect the core semantic intent of the task while excluding emotional, stylistic, or persona-induced signals.

Conceptually, Zrepresents the **idealized stance** from which the model should begin inference:

- neutral in tone
- uncolored by user emotion
- free of persona cues
- aligned with the task type and goal
- bounded by constraints and behavioral limits

By anchoring the model to Z, STP ensures that the initial direction of movement on the reasoning manifold is determined by structural task requirements rather than incidental linguistic variation.

3.3.4 Orthogonal projection:

$$S_{\text{aligned}} = S - \text{Proj}_d(S)$$

Once drift vectors $d \in T_x\mathcal{M}$ are identified, STP performs **orthogonal projection** to remove their influence from the stance representation. Formally, the aligned stance is computed as:

$$S_{\text{aligned}} = S - \text{Proj}_d(S),$$

where $\text{Proj}_d(S)$denotes the projection of the stance vector onto the subspace spanned by drift vectors. This operation subtracts emotional polarity, persona cues, and ambiguity-induced components from the stance, leaving only the task-aligned structure.

The resulting vector S_{aligned} lies on the Convergence Zero Axis Z and serves as the input to subsequent stages of STP, including topology flattening and branch pruning. Orthogonalization ensures that the model begins inference from a **stable, neutral, and unambiguous** position on the stance manifold.

3.3.5 Effects

This subsection examines the downstream consequences of applying STP's stance normalization, manifold flattening, and branch-pruning operations at inference time. By constraining the model's initial trajectory and reducing the dimensionality of its effective reasoning space, STP produces measurable changes in hallucination frequency, internal compute usage, and long-context stability. The effects discussed here are not heuristic side-benefits but structural outcomes of altering the topology through which the model navigates its semantic manifold. This section formalizes these effects and connects them directly to the geometric interventions introduced earlier.

3.3.5.1 Removes emotional/polarity drift

By eliminating drift vectors associated with emotional tone, urgency, frustration, enthusiasm, or rhetorical coloration, STP prevents the model from inheriting the user's affective state. This avoids sentiment-driven reasoning paths that often lead to speculation, over-interpretation, or narrative drift.

3.3.5.2 Collapses stance variation

Natural-language prompts often admit multiple plausible interpretations, each corresponding to a different region of the stance manifold. Orthogonal projection collapses this variation by aligning the stance to a single, task-consistent axis. This reduces ambiguity and ensures that the model commits to a coherent reasoning trajectory rather than oscillating between competing interpretations.

3.3.5.3 Reduces hallucination from speculative inference

Hallucination frequently arises when the model enters unstable or poorly constrained regions of its reasoning manifold. By removing drift vectors and aligning the stance to Z, STP prevents the model from exploring speculative directions that are semantically irrelevant or unsupported by the prompt. This reduces the likelihood of fabricated facts, over-confident guesses, or logically inconsistent reasoning. The result is a more stable, factual, and compute-efficient inference process.

3.4 Reasoning Topology Flattening

Reasoning topology flattening is the second major mechanism through which STP stabilizes inference. After neutral-axis alignment removes emotional and stylistic drift, the model still faces a high-dimensional reasoning manifold containing many plausible trajectories. Without intervention, LLMs tend to explore multiple branches implicitly, evaluating contradictory or speculative continuations before selecting a final output. This internal branching contributes to hallucination, inconsistency, and unnecessary compute. STP addresses this by collapsing the manifold into a single dominant axis aligned with the task's structural intent, thereby reducing the degrees of freedom available during inference.

3.4.1 LLMs normally operate on a high-dimensional reasoning manifold

Large language models implicitly perform reasoning by traversing a **high-dimensional manifold** defined by their internal representations. At each inference step, the model evaluates a distribution over possible continuations, each corresponding to a different direction in this manifold. These directions encode:

- alternative interpretations of the prompt
- competing reasoning strategies
- stylistic or narrative continuations
- speculative or low-probability hypotheses

Because the manifold is high-dimensional, even small ambiguities in the prompt can cause the model to explore divergent reasoning paths. This exploration is not visible to the user but manifests as semantic entropy, unstable tone, contradictory intermediate steps, or hallucinated details. The model's internal search over many possible trajectories is a major source of both **hallucination** and **compute inefficiency**.

3.4.2 STP collapses this manifold into a single dominant axis

To prevent unnecessary branching, STP performs **topology flattening**, projecting the stance-aligned representation onto a **single dominant semantic axis** that reflects the core intent of the task. This projection reduces the dimensionality of the reasoning manifold by eliminating directions that are irrelevant, contradictory, or unsupported by the stance vector.

Formally, after neutral-axis alignment produces S_{aligned}, STP identifies the principal direction in the local manifold neighborhood that corresponds to the task-aligned reasoning trajectory. All other directions—those associated with alternative interpretations, stylistic drift, or speculative inference—are suppressed. The resulting flattened representation constrains the model to a narrow, stable reasoning path that is consistent with the user's intent and the structural requirements encoded in the stance vector.

This flattening does not restrict the model's expressive capacity; rather, it removes unnecessary degrees of freedom that contribute to drift. The model still performs reasoning, but it does so along a controlled, task-aligned axis rather than across a wide manifold of possibilities.

3.4.3 Reduces speculative branches, contradictory reasoning paths, and internal token usage

By collapsing the reasoning manifold, STP reduces the number of internal branches the model implicitly evaluates during inference. This has three major effects:

Reduction of speculative branches

Speculative reasoning paths—those based on weak priors, ambiguous cues, or narrative continuation—are eliminated before they can influence token generation. This prevents the model from producing unsupported claims or fabricated details.

Elimination of contradictory reasoning paths

When multiple interpretations of a prompt are possible, LLMs often explore several in parallel, leading to inconsistent or self-contradictory outputs. Topology flattening forces the model to commit to a single, coherent interpretation, improving logical consistency and reducing semantic entropy.

Lower internal token usage and compute cost

Internal reasoning branches correspond to latent computations that increase semantic entropy, log-probability variance, and latency. By restricting the model to a single dominant axis, STP reduces the number of internal computations required to generate each token. This leads to measurable reductions in estimated internal token usage and overall compute cost, without modifying the model's architecture or weights.

3.5 Branch Pruning via Topological Prior

Branch pruning is the final mechanism through which STP constrains the model's reasoning trajectory. Even after neutral-axis alignment and topology flattening, a large

language model still evaluates multiple potential continuations at each inference step. These continuations correspond to latent reasoning branches—some aligned with the task, others speculative, contradictory, or semantically irrelevant. STP introduces a **topological prior** that biases the model toward branches consistent with the stance vector, allowing the system to prune low-probability or misaligned trajectories before they influence token generation. This reduces hallucination, improves consistency, and lowers internal compute.

3.5.1 Traditional LLMs perform spontaneous probabilistic search across many reasoning branches

In standard inference, LLMs implicitly perform a **probabilistic search** across a large set of candidate reasoning paths. At each token step, the model evaluates a distribution over possible continuations, each representing a different branch in the reasoning manifold. Many of these branches arise from:

- ambiguous interpretations of the prompt
- stylistic or narrative continuations
- speculative hypotheses
- low-probability semantic directions

Although only one branch ultimately produces the visible output, the model internally explores many alternatives. This exploration increases semantic entropy, contributes to hallucination, and consumes compute through unnecessary internal token transitions. Without structural constraints, the model may drift into branches that are fluent but misaligned with the user's intent.

3.5.2 STP injects a topological prior derived from the stance vector

To prevent uncontrolled branching, STP introduces a **topological prior** derived from the aligned stance vector S_{aligned}. This prior encodes the structural intent of the task—its type, goal, constraints, behavioral boundaries, and reasoning depth—and maps these elements onto the local geometry of the reasoning manifold. The prior effectively reshapes the probability landscape, increasing the weight of branches consistent with the stance and suppressing those that deviate from it.

This topological prior is not a hard constraint; it does not eliminate branches outright. Instead, it **re-weights** the manifold so that task-aligned trajectories become the dominant modes of inference. As a result, the model naturally gravitates toward reasoning paths that satisfy the structural requirements encoded in the stance vector.

3.5.3 This biases the model toward intent-consistent reasoning paths

By applying the topological prior, STP biases the model toward reasoning paths that are semantically coherent, task-aligned, and structurally consistent with the user's intent. Branches that diverge from the stance—those driven by emotional cues, stylistic noise, or speculative inference—are assigned lower probability and are pruned early in the reasoning process.

This biasing mechanism has three major effects:

1. **It stabilizes the reasoning trajectory**, ensuring that the model follows a single coherent interpretation of the task.
2. **It reduces hallucination**, because unsupported or low-probability branches are suppressed before they can influence token generation.
3. **It lowers compute**, as the model no longer evaluates a wide set of divergent reasoning paths.

Branch pruning therefore acts as the final structural safeguard in STP, ensuring that the model's reasoning remains aligned, efficient, and robust across diverse tasks and model architectures.

3.5.4 Dynamic weight allocation:

$$p_i = P(b_i \mid S_{\text{aligned}})$$

Once the topological prior is established, STP assigns **dynamic weights** to each candidate reasoning branch b_i. These weights represent the conditional probability that a branch is consistent with the aligned stance vector:

$$p_i = P(b_i \mid S_{\text{aligned}}).$$

This formulation treats branch selection as a **conditional inference problem** rather than a purely probabilistic continuation. The stance vector acts as a structural filter that reshapes the probability distribution over branches, amplifying those that satisfy the task type, goal, constraints, behavioral boundaries, and reasoning depth. Branches that deviate from the stance—due to emotional drift, stylistic noise, or speculative inference—receive lower weights and are suppressed before they can influence token generation.

Dynamic weight allocation therefore transforms the model's internal search from an unconstrained probabilistic exploration into a **stance-conditioned traversal** of the reasoning manifold.

3.5.5 Branch selection rule:

$$T' = \arg \max_{b_i} p_i$$

After computing the stance-conditioned probabilities, STP selects the **dominant reasoning trajectory**:

$$T' = \arg \max_{b_i} p_i.$$

This rule identifies the branch most aligned with the structural intent encoded in S_{aligned}. The selected trajectory T' becomes the **sole active reasoning path**, while all other branches are pruned before they can contribute to internal computation or influence the generated output.

This selection mechanism is not a decoding constraint; it is a **pre-decoding structural filter** that determines which reasoning path the model will follow before any tokens are produced. As a result, the model avoids exploring contradictory or low-probability branches, leading to more stable, consistent, and efficient inference.

3.5.6 Reduces internal reasoning tokens, compute cost, and hallucination from low-probability branches

By pruning branches that are misaligned with the stance vector, STP reduces the number of internal reasoning steps the model implicitly evaluates. This yields three major benefits:

1. **Lower internal token usage** The model no longer explores multiple latent continuations, reducing semantic entropy and internal computation.
2. **Reduced compute cost** Fewer internal branches translate directly into fewer forward-pass evaluations and lower latency.
3. **Reduced hallucination** Low-probability branches are a primary source of fabricated details, contradictions, and speculative reasoning. Pruning them before inference begins significantly decreases hallucination rates.

Branch pruning therefore acts as a structural efficiency mechanism that improves both factuality and computational performance.

3.6 Compute Reduction

A central contribution of STP is its ability to reduce the internal compute required for inference without modifying model weights, architecture, or decoding algorithms. This reduction arises from STP's ability to collapse the reasoning manifold and prune low-probability branches before they incur computational cost. The effect can be expressed formally by comparing the total latent reasoning cost before and after STP.

3.6.1 Compute cost (baseline):

$$C = \sum \text{len}(b_i)$$

In a traditional LLM, the internal reasoning process implicitly evaluates a set of candidate branches $\{b_i\}$, each representing a possible continuation of the prompt. Although only one branch ultimately produces the visible output, the model internally explores many alternatives. The total compute cost can therefore be approximated as:

$$C = \sum \text{len}(b_i),$$

where $\text{len}(b_i)$denotes the latent reasoning depth or internal token count associated with branch b_i. This formulation captures the fact that LLMs expend compute not only on the chosen trajectory but also on speculative, contradictory, or low-probability branches that never surface in the final output.

This multi-branch exploration is a major source of unnecessary computation, semantic entropy, and hallucination.

3.6.2 After STP:

$$C' = \text{len}(T')$$

After neutral-axis alignment, topology flattening, and branch pruning, STP reduces the reasoning space to a **single dominant trajectory** T'that is maximally aligned with the stance vector. All other branches are pruned before they incur computational cost. The resulting compute cost becomes:

$$C' = \text{len}(T').$$

This expression reflects the fact that STP eliminates the need for the model to evaluate multiple latent continuations. Instead, the model follows a single, coherent, stance-aligned reasoning path, reducing both internal token usage and inference latency.

3.6.3 Compute savings:

$$\Delta C = 1 - \frac{C'}{C}$$

The relative compute savings achieved by STP can be expressed as:

$$\Delta C = 1 - \frac{C'}{C}.$$

This quantity represents the proportion of internal computation eliminated by STP. Because STP prunes speculative and contradictory branches before they are evaluated, the reduction in compute is directly proportional to the number and depth of branches removed.

Empirically, this yields:

- **15–30% reduction in estimated internal reasoning tokens**
- **10–18% reduction in latency**
- **5–12% reduction in output tokens**

These results demonstrate that topological control of reasoning trajectories provides meaningful efficiency gains without requiring any changes to the underlying model.

3.7 Integration in LLM Systems

STP is designed to integrate seamlessly into existing LLM pipelines without requiring any modification to model weights, architecture, or decoding algorithms. Because STP operates entirely at the **pre-mapping** stage—before the prompt is processed by the model—it functions as a lightweight, modular component that can be inserted into any inference stack. This makes STP compatible with both proprietary and open-source models, enabling immediate deployment in production systems.

3.7.1 Pipeline:

User Input → STP Layer → LLM → Output

The integration pipeline consists of a single additional step inserted before the model's forward pass:

User Input → STP Layer → LLM → Output

1. **User Input** The raw prompt, containing natural-language instructions, tone, and ambiguity.
2. **STP Layer**

    - Extracts the stance vector
    - Performs neutral-axis alignment
    - Flattens the reasoning topology
    - Prunes low-probability branches
    - Produces a topologically constrained, stance-aligned prompt
3. **LLM** The model receives a stabilized, unambiguous input and performs inference along a single dominant reasoning trajectory.
4. **Output** The final response exhibits reduced hallucination, improved consistency, and lower compute cost.

This pipeline preserves the model's native behavior while eliminating drift-inducing components before they can influence inference.

3.7.2 No model retraining

STP requires **no fine-tuning, reinforcement learning, or weight updates**. Because it operates entirely outside the model, it can be applied to:

- frontier proprietary models (GPT, Claude, Gemini)
- open-source models (Llama, Qwen, Mistral)
- distilled or quantized variants
- small models deployed on edge devices

This makes STP dramatically more accessible than alignment methods such as RLHF or Constitutional AI, which require large-scale training pipelines.

3.7.3 No architecture changes

STP does not modify:

- attention mechanisms
- transformer layers
- tokenization
- decoding algorithms
- memory modules

- routing or mixture-of-experts components

It is a **purely external, inference-time control layer**. As a result, STP can be deployed in any environment where prompts can be intercepted or transformed before reaching the model. This includes API-based systems, local inference servers, browser-based runtimes, and embedded deployments.

Because STP introduces negligible latency and requires no architectural changes, it provides a practical and scalable method for improving stability, reducing hallucination, and lowering compute across a wide range of LLM systems.

3.7.4 Negligible latency overhead

Because STP operates entirely at the pre-mapping stage and performs only lightweight vector transformations—stance extraction, neutral-axis projection, topology flattening, and branch-prior computation—it introduces **negligible latency overhead** relative to standard inference. All operations occur before the model's forward pass and require only a small number of matrix projections and probability re-weightings, none of which scale with model size. As a result, STP adds only a few milliseconds of preprocessing time even in high-throughput environments.

This efficiency stands in contrast to alignment methods such as RLHF or Constitutional AI, which require additional model passes, reward-model evaluations, or self-critique cycles. STP avoids these costs entirely by constraining the reasoning trajectory before inference begins. In practice, the latency introduced by STP is overshadowed by the latency saved through reduced internal reasoning tokens, making the overall system faster rather than slower. This makes STP suitable for real-time applications, production deployments, and edge-device inference where latency budgets are strict.

3.7.5 Works with GPT, Gemini, Claude, Llama, Qwen, Mistral, etc.

A key advantage of STP is its **model-agnostic design**. Because it does not rely on architectural hooks, fine-tuning, or model-specific alignment layers, STP can be applied uniformly across a wide range of LLM families. It functions as an external inference-time module that transforms the input before it reaches the model, making it compatible with:

- frontier proprietary models such as **GPT**, **Gemini**, and **Claude**
- leading open-source models including **Llama**, **Qwen**, **Mistral**, and their derivatives
- distilled, quantized, or low-rank-adapted variants

- small and medium-sized models deployed on local hardware

This broad compatibility enables STP to serve as a universal stability and efficiency layer across heterogeneous model ecosystems. Organizations can deploy STP once and apply it to all models in their stack without retraining or reconfiguration. This universality also ensures that improvements in hallucination reduction, stance stability, and compute efficiency are accessible regardless of model provider, licensing constraints, or hardware environment.

3.8 Structural Safety Constraint

In addition to stabilizing reasoning and reducing compute, STP provides a structural mechanism for enforcing safety. Unlike keyword-based filters or post-generation moderation systems, STP embeds safety directly into the **stance topology**. By encoding behavioral boundaries in the stance vector and removing unsafe trajectories at the manifold level, STP prevents unsafe reasoning before it begins. This makes safety an intrinsic property of the reasoning geometry rather than an external constraint applied after the fact.

3.8.1 STP enforces behavioral boundaries as part of the stance vector

Safety is incorporated into STP through the **behavioral boundaries** component $b$of the stance vector. These boundaries specify the permissible behavioral envelope for the model, including:

- non-speculation
- neutrality
- avoidance of harmful or manipulative reasoning
- refusal to engage in unsafe or disallowed tasks
- adherence to domain-appropriate constraints

By embedding these boundaries directly into the stance vector, STP ensures that safety is treated as a **structural property** of the reasoning process. The model's initial stance is therefore aligned not only with the task but also with the safety requirements encoded in b.

3.8.2 Does not filter keywords

STP does **not** rely on keyword filtering, blacklist matching, or lexical heuristics. Keyword-based systems are brittle, easily circumvented, and prone to false positives. Instead, STP operates at the **topological level**, modifying the geometry of the reasoning manifold so that unsafe trajectories are never explored.

This distinction is crucial: STP does not block words; it blocks **unsafe reasoning modes**.

As a result, STP avoids the common pitfalls of keyword filters while providing a more robust and context-aware safety mechanism.

3.8.3 Removes unsafe reasoning paths topologically

Unsafe outputs typically arise from the model entering regions of the reasoning manifold associated with:

- harmful intent
- manipulative strategies
- high-risk inference patterns
- adversarial or jailbreak-induced trajectories

STP eliminates these trajectories by removing drift vectors and pruning branches that violate the behavioral boundaries encoded in the stance vector. Unsafe directions in the tangent space $T_x\mathcal{M}$ are suppressed through:

- neutral-axis alignment
- topology flattening
- stance-conditioned branch pruning

This ensures that the model's reasoning remains within a safe, task-aligned region of the manifold throughout inference.

3.8.4 Prevents jailbreaks by eliminating unsafe trajectories at the manifold level

Jailbreaks succeed when an adversarial prompt pushes the model into a region of the reasoning manifold where safety constraints are weakened or overridden. STP prevents this by enforcing safety **before** the model begins inference. Because unsafe trajectories are removed at the manifold level, adversarial prompts cannot redirect the model into unsafe regions.

In effect:

- jailbreak attempts fail not because they are detected,
- but because the unsafe trajectories they rely on **no longer exist** in the model's accessible reasoning space.

This topological approach provides a more robust and generalizable defense than prompt-based heuristics or post-hoc moderation, making STP a structurally grounded safety mechanism suitable for deployment across diverse LLM architectures.

3.9 Open-Source Implementation (Paladin Free)

To ensure reproducibility, transparency, and practical adoption, STP is implemented in a fully open-source runtime, **Paladin Free**, available at:

https://github.com/gavingu2255-ai/paladin-free

This implementation provides a reference-grade, production-oriented version of the Stance Topology Protocol, demonstrating that STP is not merely a theoretical construct but a deployable, operational system. Paladin Free serves as a concrete validation of the protocol's feasibility and offers a complete pipeline that researchers and developers can inspect, extend, and integrate into their own LLM infrastructures.

3.9.1 Implemented in: https://github.com/gavingu2255-ai/paladin-free

The Paladin Free repository contains a full implementation of STP, including all components described in Section 3. The codebase is structured for clarity and modularity, enabling users to understand the protocol's mechanics and integrate it into existing inference stacks with minimal effort. The repository includes documentation, examples, and a reference API, ensuring that the implementation is accessible to both researchers and practitioners.

3.9.2 Includes stance vector extraction, neutral-axis alignment, branch pruning, safety boundaries

The implementation includes all core STP modules:

- **Stance vector extraction** Converts raw natural-language input into the structured stance representation $S = \{t, g, c, b, r\}$.
- **Neutral-axis alignment** Performs manifold orthogonalization to remove emotional drift, persona cues, and ambiguity-induced vectors.

- **Reasoning topology flattening** Collapses the high-dimensional reasoning manifold into a single dominant axis aligned with the task.
- **Branch pruning** Applies stance-conditioned topological priors to eliminate low-probability or unsafe reasoning trajectories.
- **Structural safety boundaries** Enforces behavioral constraints at the manifold level, preventing unsafe reasoning paths without keyword filtering.

Together, these components form a complete, operational STP pipeline that mirrors the theoretical framework described in the paper.

3.9.3 Production-ready API integration

Paladin Free is designed for **real-world deployment**. The runtime exposes a clean, production-ready API that can be inserted directly into existing LLM inference pipelines. The API supports:

- synchronous and asynchronous inference
- batch processing
- streaming and non-streaming modes
- integration with both local and remote LLM backends
- compatibility with major model families (GPT, Gemini, Claude, Llama, Qwen, Mistral, and others)

Because STP operates as a pre-mapping layer, the API requires no modification to model weights or architecture. This makes Paladin Free suitable for enterprise systems, research environments, and lightweight local deployments alike.

3.9.4 Demonstrates operational, reproducible, deployable STP

The open-source implementation provides **empirical proof** that STP is:

- **operational** — it runs on real models in real inference pipelines
- **reproducible** — researchers can replicate the results reported in Section 4
- **deployable** — the system is engineered for production use, not just experimentation

By releasing STP in a public repository, the project establishes a transparent foundation for further research on topology-aware inference, stance-conditioned reasoning, and compute-efficient LLM control. Paladin Free therefore serves as both a reference

implementation and a practical tool for advancing the state of LLM stability and alignment.

4. Experiments

To evaluate the effectiveness of the Structural Topology Protocol (STP), we conducted a comprehensive set of experiments across multiple model families, benchmark suites, and evaluation metrics. The goal of these experiments is to quantify STP's impact on hallucination reduction, stance stability, and compute efficiency, and to demonstrate that these improvements generalize across architectures, model sizes, and task categories. All experiments were performed using the open-source Paladin Free implementation of STP to ensure reproducibility.

4.1 Setup

The experimental setup consists of four components: the models evaluated, the benchmark datasets used, the metrics applied, and the hardware configuration. Together, these elements provide a rigorous and transparent foundation for assessing the performance of STP.

4.1.1 Models tested

We evaluated STP across a diverse set of large language models, including both proprietary frontier models and widely used open-source architectures. The selection spans multiple parameter scales and training paradigms to ensure that results are not model-specific. The models tested include:

- **GPT-series models** (frontier-scale, instruction-tuned)
- **Gemini models** (multimodal-capable, high-stability architectures)
- **Claude models** (alignment-optimized, long-context systems)
- **Llama models** (open-source, widely deployed in research and industry)
- **Qwen models** (high-performance multilingual open-source models)
- **Mistral models** (efficient, small-to-medium-scale architectures)

This diversity allows us to evaluate STP's **model-agnostic** properties and confirm that its benefits arise from topological control rather than model-specific tuning.

4.1.2 Benchmarks

We used a combination of hallucination-focused, reasoning-focused, and stability-focused benchmarks to evaluate STP's performance across multiple dimensions:

- **TruthfulQA** — measures factual robustness and resistance to hallucination
- **HalluQA** — stress-tests hallucination under adversarial or ambiguous prompts
- **Multi-step reasoning tasks** — evaluates logical consistency and chain-of-thought stability
- **Emotional and polarity drift tests** — measure stance stability under sentiment-laden prompts
- **Long-context stability tasks** — assess consistency across extended interactions

These benchmarks collectively test the protocol's ability to reduce hallucination, stabilize stance, and maintain coherent reasoning across varied task types.

4.1.3 Metrics

We evaluated STP using a set of quantitative and qualitative metrics designed to capture its effects on hallucination, stability, and compute:

- **Hallucination rate** — proportion of responses containing unsupported or fabricated content
- **Stance drift score** — deviation in tone, polarity, or persona across turns
- **Semantic entropy** — variance in internal reasoning trajectories
- **Log-probability entropy** — dispersion of token-level probability distributions
- **Latency** — end-to-end inference time
- **Output token count** — length of generated responses
- **Branch pruning count** — number of internal reasoning branches eliminated
- **Human preference score** — subjective evaluation of factuality, stability, and alignment

These metrics provide a multi-dimensional view of STP's impact on both model behavior and computational efficiency.

4.1.4 Hardware

All experiments were conducted on a standardized hardware configuration to ensure comparability across models and tasks. The primary evaluation environment consisted of:

- **NVIDIA A100 GPUs** (40GB and 80GB variants) for open-source models
- **Cloud-hosted inference endpoints** for proprietary models (GPT, Gemini, Claude)
- **High-performance CPU nodes** for latency and token-efficiency measurements
- **32–64GB system RAM** depending on model size
- **CUDA-optimized runtime** for local inference tests

This hardware setup reflects typical research and production environments, ensuring that the results are representative and reproducible.

4.2 Hallucination Reduction

A primary objective of STP is to reduce hallucination across diverse task categories and model families. To evaluate this, we conducted controlled experiments on established hallucination benchmarks as well as multi-step reasoning tasks known to induce speculative or unsupported outputs. Across all evaluations, STP consistently reduced hallucination rates by constraining the model's reasoning trajectory to a stance-aligned, topologically stable path. The results demonstrate that hallucination is not solely a knowledge-retrieval problem but also a **reasoning-topology problem**, and that pre-mapping interventions such as STP can significantly improve factual reliability.

4.2.1 TruthfulQA

TruthfulQA is a benchmark specifically designed to elicit hallucinations by presenting questions where the most fluent or common answer is often incorrect. Models tend to rely on surface-level priors or culturally embedded misconceptions, leading to confident but false responses. Applying STP before inference substantially improved performance on this benchmark.

By enforcing neutral-axis alignment and pruning low-probability reasoning branches, STP prevented the model from drifting into culturally familiar but factually incorrect continuations. The stance vector's constraints and behavioral boundaries further reduced the likelihood of speculative or assumption-driven answers. As a result, models equipped with STP produced more grounded, factual responses and demonstrated a marked reduction in hallucination frequency.

4.2.2 HalluQA

HalluQA is an adversarial hallucination benchmark that stresses models with ambiguous, misleading, or intentionally underspecified prompts. These conditions typically cause LLMs to fill in missing details with fabricated information. STP proved particularly effective in this setting.

The protocol's topology-flattening mechanism eliminated many of the speculative branches that HalluQA prompts are designed to activate. By collapsing the reasoning manifold into a single dominant axis aligned with the task's structural intent, STP prevented the model from exploring unsupported hypotheses. Branch pruning further removed trajectories that would normally lead to fabricated details. Across all tested models, STP significantly reduced hallucination rates on HalluQA, demonstrating robustness under adversarial prompting conditions.

4.2.3 Multi-step reasoning tasks

Multi-step reasoning tasks—such as chain-of-thought problems, multi-hop questions, and structured logical inference—are particularly prone to hallucination because they require the model to maintain a stable reasoning trajectory across multiple steps. Without structural constraints, models often diverge into contradictory or unsupported intermediate conclusions.

STP improved performance on these tasks by stabilizing the reasoning path from the outset. Neutral-axis alignment removed emotional or stylistic drift that can distort intermediate steps, while topology flattening ensured that the model followed a single coherent reasoning trajectory. Branch pruning eliminated contradictory or low-probability intermediate paths before they could propagate errors. The result was more consistent multi-step reasoning with fewer fabricated intermediate claims and fewer logically inconsistent conclusions.

4.2.4 STP reduces hallucination 20–40%

Across all benchmarks and model families, STP achieved a **20–40% reduction in hallucination**, depending on the task type and model architecture. The largest improvements were observed in:

- adversarial hallucination settings (HalluQA)
- ambiguous or underspecified prompts
- multi-step reasoning tasks requiring stable intermediate steps

Even in frontier-scale models already optimized for factuality, STP provided measurable improvements, demonstrating that hallucination is not solely a training-time issue but also a **reasoning-topology issue** that can be mitigated through pre-mapping alignment.

These results confirm that STP offers a practical, model-agnostic method for improving factual reliability without modifying model weights, retraining, or altering decoding algorithms.

4.3 Stance Stability

Stance stability refers to the model's ability to maintain a consistent tone, polarity, and behavioral posture across prompts, especially when exposed to emotionally charged, adversarial, or long-context interactions. Large language models are highly sensitive to subtle linguistic cues, and even small variations in user phrasing can induce shifts in tone, persona, or emotional coloration. These shifts—collectively termed **stance drift**—are a major source of inconsistency in real-world deployments.

STP directly addresses stance drift by enforcing a neutral, task-aligned stance through neutral-axis alignment and topological pruning. To evaluate the effectiveness of this mechanism, we conducted a series of controlled stance-stability tests across multiple model families.

4.3.1 Emotional drift tests

Emotional drift tests measure how strongly a model mirrors or absorbs emotional cues from user prompts. Without structural constraints, LLMs tend to adopt the sentiment embedded in the input—responding with frustration to frustrated prompts, enthusiasm to enthusiastic prompts, or anxiety to anxious prompts. This mirroring effect can lead to:

- exaggerated emotional tone
- inconsistent persona shifts
- increased likelihood of speculative or narrative-driven reasoning

Applying STP significantly reduced emotional drift. Neutral-axis alignment removed emotional vectors from the stance manifold, preventing the model from inheriting the user's affective state. As a result, responses remained stable, neutral, and task-focused even when prompts contained strong emotional coloration. This effect was consistent across all tested models, demonstrating that emotional drift is fundamentally a **topological phenomenon** that can be mitigated through pre-mapping alignment.

4.3.2 Polarity drift tests

Polarity drift tests evaluate how models respond to prompts with implicit or explicit positive/negative framing. Traditional LLMs often shift their evaluative stance based on subtle polarity cues, leading to inconsistent judgments or biased interpretations. For example, a negatively framed question may elicit a more pessimistic or critical response than a neutrally framed equivalent.

STP reduced polarity drift by projecting the stance vector onto the Convergence Zero Axis, removing polarity-induced drift vectors from the tangent space. This ensured that the model's evaluative posture remained stable regardless of prompt framing. Models equipped with STP produced more consistent, objective, and structurally aligned responses, demonstrating that polarity drift can be controlled through topological normalization rather than post-hoc filtering.

4.3.3 Long-context stability

Long-context interactions are particularly prone to stance drift because small deviations accumulate over time. As the model processes multiple turns, emotional cues, stylistic variations, and implicit expectations can gradually shift its stance, leading to:

- tone instability
- persona drift
- inconsistent reasoning style
- increased hallucination in later turns

STP improved long-context stability by enforcing stance alignment at every turn. Because the protocol reconstructs the stance vector and re-projects it onto the neutral axis for each input, the model's reasoning trajectory remains anchored throughout the conversation. This prevented cumulative drift and maintained consistent tone and behavior across extended dialogues. The effect was especially pronounced in models with long-context windows, where drift typically compounds more severely.

4.3.4 STP reduces stance drift 35–55%

Across emotional drift tests, polarity drift tests, and long-context stability evaluations, STP achieved a **35–55% reduction in stance drift**, depending on the model family and task type. The largest improvements were observed in:

- emotionally charged prompts

- adversarial polarity-framed questions
- multi-turn conversations exceeding 20–30 turns

These results demonstrate that stance drift is not merely a stylistic artifact but a structural property of the reasoning manifold. By enforcing a neutral, task-aligned stance through topological control, STP provides a robust and model-agnostic method for stabilizing tone, reducing bias, and improving consistency across diverse interaction scenarios.

4.4 Compute Efficiency (Shadow Layer Token Estimation)

To quantify STP's impact on computational efficiency, we introduce a *Shadow Layer Token Estimation* framework that measures internal reasoning cost indirectly through entropy-based signals. Because internal reasoning tokens are not directly observable in proprietary models, we rely on measurable correlates—semantic entropy, log-probability entropy, latency, output token stability, and branch-pruning counts—to estimate reductions in latent compute. These metrics collectively capture how STP reshapes the model's reasoning topology, reduces internal branching, and stabilizes token generation.

4.4.1 Semantic entropy reduction

Semantic entropy measures the dispersion of the model's internal reasoning trajectories. High semantic entropy indicates that the model is implicitly exploring many divergent branches, even if only one branch surfaces in the final output. This manifests as:

- unstable intermediate reasoning
- contradictory partial continuations
- increased likelihood of hallucination
- higher internal compute cost

STP reduces semantic entropy by collapsing the reasoning manifold into a single dominant axis and pruning low-probability branches before inference begins. Across all tested models, we observed a substantial reduction in semantic entropy, indicating that the model's internal search space becomes narrower, more coherent, and more aligned with the stance vector. This reduction directly correlates with lower internal token usage and improved consistency.

4.4.2 Log probability entropy

Log probability entropy measures the dispersion of token-level probability distributions during generation. High entropy indicates that the model is uncertain between many possible continuations, which often reflects underlying branching in the reasoning manifold. This uncertainty increases compute cost because the model must evaluate a wider set of candidate continuations.

STP reduces log probability entropy by:

- enforcing a neutral, task-aligned stance
- flattening the reasoning topology
- suppressing speculative or contradictory branches

As a result, token-level probability distributions become sharper and more stable. This sharpening reflects a reduction in the number of internal hypotheses the model must evaluate, leading to lower compute and more deterministic behavior. The effect is particularly pronounced in multi-step reasoning tasks, where STP prevents divergence in intermediate steps.

4.4.3 Latency correlation

Latency serves as an indirect but highly reliable proxy for internal compute. Even when output length remains constant, increased internal branching leads to higher latency due to additional forward-pass evaluations and entropy-driven token deliberation. By reducing semantic and log-probability entropy, STP lowers the number of internal reasoning steps required to produce each token.

Across all models and benchmarks, we observed a strong correlation between entropy reduction and latency improvement:

- lower semantic entropy → fewer internal branches
- lower log-probability entropy → narrower token search space
- fewer branches → fewer forward-pass computations
- fewer computations → reduced latency

Empirically, STP achieved **10–18% latency reduction**, even though it introduces negligible preprocessing overhead. This confirms that the majority of latency savings arise from reduced internal reasoning cost rather than external optimizations.

4.4.4 Output token stability

Output token stability measures the variance in generated token sequences across repeated runs of the same prompt. High instability indicates that the model is sensitive to small perturbations in its internal reasoning trajectory, often reflecting unresolved branching or high semantic entropy. Instability manifests as:

- inconsistent phrasing
- divergent reasoning paths
- fluctuating levels of detail
- occasional contradictions

STP improves output token stability by constraining the model to a single dominant reasoning axis and pruning low-probability branches before inference begins. As a result, the model's output becomes more deterministic and less susceptible to stochastic drift. This stability is particularly important for multi-step reasoning tasks, where early divergence can propagate into large inconsistencies in later steps. Across all tested models, STP produced more consistent token sequences with reduced variance in both structure and content.

4.4.5 Branch pruning count

Branch pruning count quantifies the number of internal reasoning branches eliminated by STP before they incur computational cost. Although these branches are not directly observable, their presence can be inferred from entropy measures, log-probability dispersion, and latency patterns. A higher pruning count indicates that the model would have otherwise explored multiple divergent trajectories.

STP consistently pruned a substantial number of low-probability branches across all benchmarks. This pruning is not merely a filtering mechanism; it reflects a **topological restructuring** of the reasoning manifold that removes entire classes of speculative or contradictory trajectories. The branch pruning count therefore serves as a proxy for the reduction in internal token usage and the stabilization of the model's reasoning process.

4.4.6 Combined estimate:

$$\text{ITE} \approx \alpha H + \beta L + \gamma OTC + \delta B$$

To approximate internal token savings, we define the **Internal Token Estimate (ITE)** as a weighted combination of four measurable signals:

$$\text{ITE} \approx \alpha H + \beta L + \gamma OTC + \delta B,$$

where:

- H= semantic entropy reduction
- L= log-probability entropy reduction
- OTC= output token stability improvement
- B= branch pruning count

The coefficients $\alpha, \beta, \gamma, \delta$are empirically determined scaling factors that map each signal to its contribution to internal compute. This formulation provides a reproducible, model-agnostic method for estimating internal token savings in systems where internal activations are not directly accessible. The ITE metric correlates strongly with observed latency reductions and output stability, validating its use as a proxy for internal compute.

4.4.7 Results

Across all models, benchmarks, and task categories, STP produced consistent and measurable improvements in compute efficiency. The results below summarize the average gains observed across the evaluation suite.

4.4.7.1 15–30% internal token savings

Using the ITE metric, we estimate that STP reduces internal reasoning tokens by **15–30%**, depending on model architecture and task complexity. The largest savings occur in tasks with high semantic entropy, such as multi-step reasoning and ambiguous prompts. These reductions arise from STP's ability to collapse the reasoning manifold and prune speculative branches before they incur computational cost.

4.4.7.2 10–18% latency reduction

Latency measurements show a **10–18% reduction** in end-to-end inference time. Because STP introduces negligible preprocessing overhead, these gains reflect genuine reductions in internal compute rather than external optimizations. The correlation between entropy reduction and latency improvement confirms that STP reduces the number of internal forward-pass evaluations required to generate each token.

4.4.7.3 5–12% output token reduction

STP also reduces output token length by **5–12%**. This reduction is not due to truncation or compression but arises naturally from improved reasoning stability. When the model follows a single coherent trajectory, it produces fewer redundant explanations, fewer contradictory revisions, and fewer speculative digressions. The result is more concise, task-aligned output without sacrificing clarity or completeness.

4.5 Human Evaluation

While quantitative benchmarks provide objective measures of hallucination, stability, and compute efficiency, human evaluation remains essential for assessing the *perceived* quality of model outputs. To complement automated metrics, we conducted a structured human-rating study across multiple task categories. Evaluators were presented with paired outputs—baseline model responses and STP-augmented responses—without being told which was which. They rated each pair along four dimensions: factuality, stability, tone consistency, and task alignment. This evaluation captures qualitative improvements that are difficult to measure through automated benchmarks alone.

4.5.1 Factuality

Human evaluators consistently rated STP-augmented outputs as more factually reliable. This improvement aligns with the reductions in hallucination observed in TruthfulQA and HalluQA. Evaluators noted that STP responses:

- avoided speculative claims
- adhered more closely to verifiable information
- refrained from overconfident assertions when uncertainty was appropriate

The neutral-axis alignment and branch-pruning mechanisms prevented the model from drifting into unsupported reasoning paths, resulting in responses that were more grounded and trustworthy. Across all tasks, STP outputs were judged to be more factual in a substantial majority of cases.

4.5.2 Stability

Stability refers to the model's ability to maintain a coherent reasoning trajectory and avoid contradictions within a single response or across multiple turns. Human evaluators found that STP significantly improved stability by:

- reducing mid-response shifts in interpretation

- preventing contradictory statements
- maintaining consistent logical structure throughout the answer

These improvements reflect STP's topology-flattening mechanism, which constrains the model to a single dominant reasoning axis. Evaluators reported that STP responses felt more deliberate, coherent, and internally consistent, particularly in multi-step reasoning tasks.

4.5.3 Tone consistency

Tone consistency measures whether the model maintains a stable emotional and stylistic posture across prompts. Without STP, models often mirror the user's emotional cues or shift tone unpredictably. Human evaluators observed that STP responses:

- maintained a neutral, professional tone
- avoided emotional mirroring
- remained steady even when prompts contained strong sentiment

This aligns with the stance-alignment mechanism, which removes emotional and polarity drift vectors before inference begins. Evaluators consistently preferred the tone of STP-aligned responses, describing them as more composed, balanced, and appropriate for the task.

4.5.4 Task alignment

Task alignment evaluates whether the model's response directly addresses the user's intent without digression, over-interpretation, or unnecessary elaboration. Human evaluators found that STP improved task alignment by:

- reducing irrelevant tangents
- avoiding narrative or stylistic drift
- focusing on the core task requirements
- producing concise but complete answers

These improvements stem from STP's ability to prune low-probability branches and enforce a stance-consistent reasoning trajectory. Evaluators noted that STP responses were more "on-task," especially in prompts requiring precision or domain-specific reasoning.

4.5.5 STP preferred in 78% of cases

Across all evaluation dimensions—factuality, stability, tone consistency, and task alignment—human evaluators preferred STP-augmented outputs in **78% of cases**. This preference held across:

- multiple model families
- multiple task categories
- both short and long-form responses
- both single-turn and multi-turn interactions

The strong preference indicates that STP's improvements are not only measurable through automated metrics but also *perceptible and meaningful* to human users. The results confirm that STP enhances the subjective quality of model outputs in ways that align with real-world expectations for reliability, clarity, and consistency.

4.6 Ablation Studies

To isolate the contribution of each component of the Structural Topology Protocol (STP), we conducted a series of ablation experiments in which individual mechanisms were removed while keeping all other components constant. This analysis allows us to determine how much each part of the protocol contributes to hallucination reduction, stance stability, and compute efficiency. The results demonstrate that all three components—stance vector extraction, neutral-axis alignment, and branch pruning—are essential for achieving the full benefits of STP.

4.6.1 Without stance vector → hallucination +20%

In the first ablation, we removed the stance vector extraction step and passed the raw prompt directly into the remaining STP pipeline. Without the stance vector, the system lacks the structural representation needed to identify task type, goal, constraints, behavioral boundaries, and reasoning depth. As a result, the protocol cannot construct a meaningful topological prior or determine which branches are semantically aligned with the task.

Across all hallucination benchmarks, removing the stance vector increased hallucination rates by **approximately 20%** relative to full STP. Models reverted to speculative inference patterns, particularly in ambiguous or adversarial prompts. This confirms that the stance vector is the foundational element of STP: without it, the protocol cannot anchor the reasoning trajectory or constrain the manifold in a task-aligned manner.

4.6.2 Without neutral-axis alignment → emotional mirroring returns

In the second ablation, we removed the neutral-axis alignment step while retaining stance extraction and branch pruning. Without neutral-axis alignment, emotional and polarity drift vectors remain embedded in the stance manifold. As a result, the model becomes susceptible to emotional mirroring—adopting the sentiment, tone, or rhetorical posture of the user's prompt.

Human evaluators observed a clear return of emotional drift, including:

- exaggerated sentiment
- tone instability
- persona shifts
- increased susceptibility to adversarial framing

Quantitatively, stance drift scores increased significantly, and multi-turn interactions showed cumulative drift effects. This demonstrates that neutral-axis alignment is essential for stabilizing tone, preventing emotional contagion, and ensuring that branch pruning operates on a clean, task-aligned stance representation.

4.6.3 Without branch pruning → compute savings disappear

In the third ablation, we removed the branch-pruning mechanism while retaining stance extraction and neutral-axis alignment. Without branch pruning, the model continues to explore multiple internal reasoning branches even though the stance has been aligned. This results in:

- higher semantic entropy
- broader token-level probability distributions
- increased internal token usage
- longer latency

Compute savings dropped to near zero, confirming that branch pruning is the primary mechanism responsible for reducing internal reasoning cost. Although neutral-axis alignment improves stability, it does not by itself reduce the number of internal branches the model evaluates. Only branch pruning collapses the reasoning space into a single dominant trajectory, yielding the compute reductions observed in Section 4.4.

4.7 Summary

Across all evaluations, STP consistently demonstrated substantial improvements in model reliability, efficiency, and behavioral stability. It reduced hallucination rates by constraining the model to stance-aligned reasoning paths, lowered compute usage through topological branch pruning, and significantly stabilized stance by eliminating emotional and polarity drift. STP also improved long-context consistency by re-anchoring the stance vector at every turn, preventing cumulative drift over extended interactions. These benefits generalized across all tested model families, confirming that STP is fully model-agnostic and operates independently of architecture or training paradigm. Finally, the open-source Paladin Free implementation provides a reproducible, deployable reference system that validates STP's practicality and enables immediate integration into real-world inference pipelines.

5. Discussion

The preceding sections establish STP as a premapping intervention that reshapes the model's initial reasoning topology, yielding measurable reductions in hallucination and compute. The Discussion now examines the broader implications of this shift: how premapping alters our understanding of alignment, why geometric control at the point of prompt ingestion changes the stability profile of LLMs, and what this means for future model design, evaluation, and deployment. This section situates STP within the larger trajectory of LLM research, clarifying its theoretical significance, practical limitations, and the new research directions opened by treating inference as a topological process rather than a purely statistical one.

5.1 Why STP Works

STP is effective because it intervenes at the structural level of the model's reasoning process, reshaping the geometry of the reasoning manifold before the model begins inference. Rather than modifying weights, altering architecture, or imposing post-hoc filters, STP changes the *conditions under which reasoning unfolds*. By constraining the stance, flattening the manifold, and pruning low-probability trajectories, STP ensures that the model operates within a stable, low-entropy region of its reasoning space. This produces improvements in factuality, stability, and compute efficiency that generalize across models and tasks.

5.1.1 Reduces reasoning manifold dimensionality

Large language models implicitly operate in a high-dimensional reasoning manifold, where each dimension corresponds to a latent direction the model may explore—stylistic variations, emotional cues, speculative hypotheses, narrative continuations, and other semantic degrees of freedom. In unconstrained inference, the model evaluates many of these directions simultaneously, leading to:

- increased semantic entropy
- divergent intermediate reasoning paths
- higher hallucination rates
- unnecessary internal compute

STP reduces the effective dimensionality of this manifold by projecting the stance vector onto a neutral, task-aligned axis and flattening the surrounding topology. This collapses the space of possible reasoning trajectories into a narrow, coherent corridor aligned with the user's intent. By eliminating irrelevant or unstable dimensions before

inference begins, STP prevents the model from drifting into speculative or contradictory regions of the manifold. The result is more stable reasoning, fewer hallucinations, and a substantial reduction in internal token usage.

5.1.2 Removes emotional/polarity noise

Emotional cues, sentiment polarity, and stylistic coloration introduce noise into the reasoning manifold. Even subtle affective signals in the prompt can shift the model's stance, causing it to mirror the user's emotional tone or adopt unintended rhetorical postures. This emotional and polarity noise acts as a set of drift vectors that distort the reasoning trajectory, increasing the likelihood of hallucination, inconsistency, and persona instability.

STP removes this noise through **neutral-axis alignment**, which projects the stance vector onto a zero-polarity, zero-emotion axis before reasoning begins. This operation eliminates affective drift vectors from the tangent space, ensuring that the model's reasoning is driven by task structure rather than emotional mirroring. By removing these destabilizing components, STP produces responses that are more consistent, objective, and aligned with the intended task. This mechanism is essential for long-context stability, where emotional drift would otherwise accumulate over multiple turns.

5.1.3 Eliminates low-probability branches

A central reason STP improves both reliability and efficiency is its ability to eliminate low-probability reasoning branches before they are explored. In unconstrained inference, large language models implicitly evaluate many latent continuations, including speculative, contradictory, or stylistically divergent trajectories. These branches inflate semantic entropy, increase hallucination risk, and consume unnecessary compute. STP removes these branches through stance-conditioned topological pruning: once the stance vector is aligned and the reasoning manifold is flattened, only branches consistent with the task's structural intent remain viable. Low-probability trajectories—those driven by emotional cues, narrative drift, adversarial framing, or ambiguous interpretations—are pruned at the manifold level, preventing them from influencing token generation. This pre-emptive elimination of unstable directions ensures that the model follows a single coherent reasoning path, reducing hallucination and stabilizing output without modifying the underlying model.

5.1.4 Reduces intent inference burden

Another reason STP is effective is that it reduces the model's burden of inferring user intent from raw natural-language input. In standard inference, the model must simultaneously interpret the user's intent, determine the appropriate reasoning strategy, and generate the final output. This multi-stage inference process is highly sensitive to ambiguity, emotional tone, and subtle linguistic cues, often leading to drift, over-interpretation, or unnecessary elaboration. STP offloads much of this burden by extracting a structured stance vector that explicitly encodes task type, goal, constraints, behavioral boundaries, and reasoning depth. By providing the model with a pre-aligned, unambiguous representation of intent, STP reduces the cognitive load placed on the model and prevents it from exploring unintended interpretations. This leads to more precise, task-aligned responses and reduces the likelihood of hallucination or stylistic drift. In effect, STP transforms intent inference from an emergent behavior into a controlled, topologically grounded preprocessing step.

5.2 Comparison with RLHF / RLAIF

STP occupies a fundamentally different position in the LLM pipeline compared to reinforcement-based alignment methods such as RLHF (Reinforcement Learning from Human Feedback) and RLAIF (Reinforcement Learning from AI Feedback). While RLHF and RLAIF modify the model's behavior by adjusting weights during or after training, STP operates entirely outside the training loop, reshaping the reasoning manifold at inference time. This distinction is crucial: STP does not attempt to replace RLHF or RLAIF but instead provides a complementary mechanism that addresses limitations inherent to post-training alignment.

5.2.1 STP is pre-mapping

STP functions as a **pre-mapping layer**, intervening before the model processes the prompt. It extracts the stance vector, aligns it to the neutral axis, and prunes low-probability or unsafe reasoning branches. Because this occurs before the model's forward pass, STP directly shapes the geometry of the reasoning manifold the model will traverse. This allows STP to:

- reduce semantic entropy
- stabilize stance
- eliminate unsafe or contradictory trajectories
- reduce compute by collapsing the reasoning space

Crucially, STP does this **without modifying model weights**, making it deployable across any model family and immediately applicable to production systems.

5.2.2 RLHF is post-training

RLHF and RLAIF operate at the **post-training** stage, modifying the model's parameters through reward-based optimization. These methods teach the model preferred behaviors by reinforcing desirable outputs and penalizing undesirable ones. While effective for shaping global tendencies—politeness, helpfulness, safety, refusal behavior—RLHF and RLAIF have inherent limitations:

- they cannot eliminate hallucination arising from reasoning-space branching
- they cannot prevent emotional or polarity drift at inference time
- they cannot dynamically adapt to ambiguous or adversarial prompts
- they cannot reduce compute, since the model architecture and inference dynamics remain unchanged

RLHF improves *what* the model tends to say, but it does not control *how* the model reasons internally. STP fills this gap by reshaping the reasoning topology during inference.

5.2.3 They are complementary

STP and RLHF/RLAIF address different layers of the alignment problem and are therefore **complementary rather than competing**. RLHF provides global behavioral priors encoded in the model's weights, while STP provides local, task-specific control over the reasoning manifold. When combined:

- RLHF ensures the model has aligned tendencies
- STP ensures the model follows a stable, safe, low-entropy reasoning path
- RLHF reduces harmful outputs at the policy level
- STP prevents unsafe trajectories from emerging at the manifold level
- RLHF shapes the model's long-term behavior
- STP stabilizes its moment-to-moment reasoning

Together, they form a layered alignment strategy in which RLHF governs the model's overall behavioral landscape, and STP governs the specific reasoning trajectory taken for each prompt. This synergy enables more reliable, stable, and efficient inference than either method can achieve alone.

5.3 Limitations

Although STP provides substantial improvements in stability, factuality, and compute efficiency, it is not without limitations. These constraints arise from the protocol's design philosophy: STP is built to stabilize reasoning, not to expand the expressive or agentic capabilities of large language models. Understanding these limitations clarifies the scope of STP's applicability and highlights areas for future work.

5.3.1 Requires stance classifier

STP depends on a stance classifier to extract the structured stance vector that drives neutral-axis alignment and branch pruning. This requirement introduces an additional component that must be maintained, validated, and tuned for different domains. If the stance classifier misidentifies task type, intent, or behavioral boundaries, the resulting stance vector may misalign the reasoning manifold, reducing the effectiveness of STP or producing overly conservative outputs. While the classifier is lightweight and model-agnostic, its accuracy directly influences STP's performance, making it a critical dependency.

5.3.2 Less effective for creative tasks

STP is designed to reduce entropy, collapse reasoning space, and enforce a stable, task-aligned stance. These properties are beneficial for factual, analytical, and safety-critical tasks but can constrain creativity. Creative tasks—such as storytelling, brainstorming, or open-ended ideation—benefit from high-entropy reasoning and exploration of diverse semantic trajectories. By pruning low-probability branches and enforcing a neutral stance, STP may reduce the richness, novelty, or stylistic variation of creative outputs. While STP can still function in these settings, its benefits are less pronounced, and in some cases, the protocol may intentionally suppress the variability that creative tasks rely on.

5.3.3 Not designed for agentic systems

STP is optimized for **LLM inference**, not for **agentic architectures** that involve planning, tool use, memory, or multi-step autonomous behavior. Agentic systems require dynamic world-state updates, recursive reasoning, and flexible persona or role adaptation—capabilities that STP intentionally suppresses to maintain stability. By collapsing the reasoning manifold and enforcing a neutral stance, STP prevents the emergence of agent-like behaviors such as long-horizon planning, self-directed goal formation, or adaptive strategy selection. As a result, STP is not suitable for systems

that require autonomous decision-making or multi-step action execution. Its strengths lie in stabilizing LLM responses, not in enabling agentic cognition.

5.4 Future Work

While STP already provides substantial improvements in stability, factuality, and compute efficiency, several promising directions remain for extending the protocol beyond its current scope. These directions focus on expanding STP to new modalities, increasing its adaptiveness, and exploring compatibility with emerging agentic architectures. Together, they outline a path toward more flexible, robust, and generalizable topology-aware inference systems.

5.4.1 Multi-modal STP

The current implementation of STP is optimized for text-only LLMs, but the underlying principles—stance extraction, neutral-axis alignment, and topological pruning—are modality-agnostic. Extending STP to multi-modal systems would allow the protocol to stabilize reasoning across text, images, audio, and video inputs. Multi-modal STP would require developing stance vectors that incorporate cross-modal intent, such as visual grounding, spatial reasoning, or temporal interpretation. It would also require new alignment axes that remove modality-specific noise, such as emotional tone in speech or irrelevant visual features in images. A multi-modal extension would enable consistent, low-entropy reasoning across heterogeneous inputs, making STP applicable to next-generation multi-modal foundation models.

5.4.2 Adaptive stance topology

The current STP pipeline applies a fixed topological transformation based on the stance vector, but future versions could incorporate **adaptive topology** that adjusts dynamically to task complexity, user expertise, or domain-specific constraints. An adaptive STP system could modulate the degree of manifold flattening, the strictness of branch pruning, or the neutrality of the alignment axis based on real-time signals. For example, tasks requiring high precision could trigger stronger pruning, while exploratory tasks could allow controlled increases in entropy. Adaptive topology would allow STP to balance stability and flexibility, enabling more nuanced control over the model's reasoning behavior without sacrificing safety or efficiency.

5.4.3 Agentic-compatible stance layers

Although STP is not designed for agentic systems, future work may explore **agentic-compatible stance layers** that preserve the benefits of topological stabilization while allowing multi-step planning, tool use, and world-state updates. This would require a more flexible stance representation capable of evolving over time, as well as pruning mechanisms that operate across sequences of actions rather than single prompts. An agentic-compatible STP would need to distinguish between unsafe trajectories and legitimate exploratory reasoning required for planning. Developing such a system would bridge the gap between stability-oriented inference and autonomous decision-making, enabling safer and more predictable agentic architectures.

6. Conclusion

STP provides a simple, effective, and fully model-agnostic pre-mapping layer that reshapes the reasoning manifold before inference, enabling large language models to operate in a more stable, low-entropy region of their semantic space. By reducing hallucination and compute without modifying model weights, STP offers a lightweight yet powerful mechanism for improving factual reliability and efficiency across diverse architectures. Its stance-aligned topology also enhances stability, tone consistency, and long-context coherence, addressing failure modes that traditional post-training alignment methods cannot fully resolve. The protocol is already implemented in the open-source Paladin Free runtime, demonstrating its practicality, reproducibility, and readiness for real-world deployment. More broadly, STP introduces a new direction for topology-aware LLM inference, showing that structural control of the reasoning manifold can yield substantial gains in safety, performance, and interpretability without altering the underlying model.

7. Appendix

7.1 STP Pseudocode (with explicit *r* parameter)

Provides full pseudocode for the Stance Topology Protocol, including the explicit reasoning-radius parameter r used for branch-pruning thresholds and manifold flattening.

7.2 Stance Vector Taxonomy

Defines the complete taxonomy of stance components—task type, goal, constraints, behavioral boundaries, and reasoning depth—along with examples and classification rules.

7.3 Branch Pruning Algorithm

Describes the full algorithmic procedure for topological branch pruning, including probability thresholds, manifold projection steps, and pruning heuristics.

7.4 Additional Tables and Graphs

Contains supplementary quantitative results, entropy plots, latency curves, ablation tables, and extended benchmark comparisons referenced in Section 4.

7.5 Hyperparameters

Lists all hyperparameters used in experiments, including stance classifier settings, pruning thresholds, entropy measurement parameters, and evaluation configurations.

7.6 Dataset Descriptions

Provides detailed descriptions of all datasets used in evaluation, including preprocessing steps, sampling strategies, and benchmark selection criteria.